

BUSINESS CASE 1 – SEGMENTING CLIENTS

INTRODUCTION

In the financial industry, understanding customers is a key driver for delivering personalized services and improving business decisions. In this project, we address this limitation by adopting a data-driven approach to client segmentation. Our goal is to identify meaningful groups of personas by using unsupervised machine learning techniques on heterogeneous data. This allows us to move from simplistic classifications to more realistic and actionable customer profiles, supporting more effective financial strategies.

DATA OVERVIEW & PREPROCESSING

We loaded the dataset into a pandas DataFrame, performed an initial inspection, and removed the ID column as it does not provide useful information for clustering. The data reveals a heterogeneous structure, combining demographic, financial, behavioral, and categorical variables, which requires a differentiated preprocessing approach. To address this, we grouped the variables into categorical features (e.g., Gender, Job, Area), demographic numerical variables (Age, FamilySize), and financial/behavioral variables, the latter already expressed in normalized form within the $[0,1]$ range. We then applied a Scaling to the demographic variables to ensure comparability across features, while leaving financial variables unchanged to preserve their original interpretation. Categorical variables were transformed using one-hot encoding, allowing us to include them in the analysis without introducing redundancy in binary cases. Finally, all processed features were combined into a single feature matrix, and we verified data quality through checks on data types, missing values, and summary statistics, confirming a clean and consistent dataset ready for clustering.

As a preliminary exploratory step, we also analyzed variable distributions using histograms and boxplots for numerical features (such as Debt, ESG, Digital, Income, Wealth) and pie charts for categorical variables (e.g., Gender, Area, Job). Overall, the variables show relatively balanced and well-behaved distributions, with no significant outliers or skewness issues. One notable insight is the strong concentration of clients in the “Employee/Worker” category and in the “North” area, which could be relevant when interpreting the final clusters.

DISTANCES

We defined three distance measures to handle mixed data.

- **Gower distance** that uses Hamming for categorical, normalized absolute difference for numerical, and then averaged for the number of total variables:

$$D_{Gower}(i, j) = \frac{1}{p} \sum_{k=1}^p d_k(i, j)$$

This is the standard reference for heterogeneous data.

- **Block-based custom distance:** variables are grouped (nominal, ordinal, demographic, financial), each group has a weight (0.05,0.05,0.10,0.80), and within each group we compute a distance (Hamming, rank-normalized Manhattan, Robust-scaled Manhattan, weighted Manhattan). The final distance is $D(i,j)=0.05D_{nom}+0.05D_{ord}+0.10D_{demo}+0.80D_{fin}$. Here, each D is already an average over the variables in that block, so information is compressed at group level.

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- **Feature-level weighted distance (Gower-like):** each variable contributes directly with a specific weight w_k , without grouping:

$$D(i, j) = \sum_k w_k d_k(i, j)$$

where ordinal variables use normalized ranks $\frac{|r_i - r_j|}{\text{range}}$, numerical variables use scaled differences, and financial variables use weighted Manhattan terms. This is effectively a **weighted Gower distance**, since it preserves the same normalization logic but assigns different importance to each feature, avoiding loss of information due to block averaging.

PCA- BASED OVERVIEW OF DATA STRUCTURE

To gain a clearer overall picture of the data structure, we applied Principal Component Analysis (PCA), a linear dimensionality reduction technique that projects data onto orthogonal directions maximizing variance, based on the Euclidean structure of the feature space. The first three components explain only about 47% of the total variance, indicating that the dataset is not well captured in a low-dimensional linear space. Consistently, both 2D and 3D embeddings show no clear cluster separation, suggesting that PCA is not suitable to reveal meaningful structure in this heterogeneous dataset.

FACTOR ANALYSIS and INDEPENDENT COMPONENT ANALYSIS

To further investigate the underlying structure of the data, we applied Factor Analysis (FA), a covariance-based method that represents each variable as a combination of a few latent factors. The results show that Factor 1 is clearly dominated by financial and

behavioral variables (e.g., Digital -0.75, Luxury -0.74, LifeStyle -0.73, FinEdu -0.66, Income -0.62, Wealth -0.59), suggesting the presence of a strong underlying “economic/behavioral profile”. In contrast, Factors 2 and 3 exhibit weaker and less consistent loadings, making their interpretation less clear. This is confirmed by the low mean communality (0.32), indicating that only a limited portion of the variance is explained, with several variables (e.g., Gender, Area, Investments) poorly represented.

We then extended the analysis using Independent Component Analysis (ICA), which aims to identify statistically independent components rather than merely uncorrelated ones. However, the resulting embedding does not reveal clear or meaningful structures, showing instead fragmented and geometrically separated patterns that are difficult to interpret in a clustering context.

Overall, FA proves useful for identifying a dominant economic dimension and highlighting its driving variables, while ICA, just like PCA, does not provide additional meaningful insight. Both methods are therefore limited in capturing the full structure of the dataset, confirming that the data cannot be effectively summarized by a small number of latent components and motivating the use of more flexible, custom distance-based approaches for clustering.

t-SNE ANALYSIS FOR DISTANCE EVALUATION

We used t-SNE to project high-dimensional data into 2D and visually assess how well different distance measures capture similarity between observations. In the first experiment, t-SNE was applied to the OHE dataset using standard metrics (Euclidean, Manhattan, Cosine): across all tested perplexity values (5–100), the embeddings remain diffuse with weak separation, indicating that these distances fail to capture meaningful patterns in heterogeneous data. In the second experiment, we used the weighted custom distance: in this case, the embeddings show a more structured and coherent organization of the data, particularly for perplexity values from 30 to 45, with some emerging cluster patterns that are more discernible compared to standard distances. Although the three clusters are not sharply separated, the custom weighted distance provides a clearer and more informative representation of similarities.

UMAP: NONLINEAR EMBEDDING AND DISTANCE IMPACT

Similar to t-SNE but designed to be more stable and to preserve part of the global structure, UMAP was applied to obtain a more reliable low-dimensional representation of the data.

Using standard distance metrics on the original feature space, the 2D embeddings show fragmented and scattered structures, with several small groups but no clear global separation, suggesting that similarities are only partially captured.

In contrast, when UMAP is applied using the precomputed custom, the resulting embedding displays more compact and better-separated clusters, indicating a more coherent representation of the underlying data structure. The separation between the three clusters became even more evident when we moved to a 3D setting and applied the UMAP again.

Overall, while standard distances highlight some local patterns, the custom distance improves interpretability, providing a clearer and more meaningful organization of the data, and it gave us reason to believe the best number of clusters to divide our data into might range from 2 to 6.

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PACMAP: DIFFERENCES BETWEEN DISTANCES

We further applied PaCMAP, a nonlinear embedding method designed to preserve both local neighborhoods and mid-range structure, providing a balance between t-SNE and UMAP.

Using the standard feature matrix X with Euclidean geometry, the 2D embedding appears fragmented, with several small, disconnected groups and one denser region, indicating the presence of local similarities but no clear global organization. The comparison across alternative metrics (Manhattan, cosine, correlation) confirms that embeddings remain unstable and fragmented when standard distances are used, with results strongly dependent on the chosen metric and lacking consistent structure.

When PaCMAP is instead applied using the custom distance matrix, the structure becomes significantly clearer: clusters are more compact, better separated, and more coherent, suggesting that the distance better captures the true similarities between observations. The use of a smaller neighborhood parameter ($n_neighbors=5$) further emphasizes local structure, enhancing cluster definition.

K-MEDOIDS CLUSTERING ON CUSTOM DISTANCE

To further investigate the intrinsic structure of the data, we applied K-medoids clustering using the customized distance matrix. Unlike K-means, K-medoids selects real

observations (medoids) as cluster representatives, making it more robust to non-Euclidean distances and outliers.

Across the 3D embeddings ($k = 2-6$), a consistent stratified structure emerges: clusters tend to organize along a vertical gradient, with a relatively stable separation between lower (blue/purple) and upper (yellow/orange) regions. However, as k increases, clusters become progressively fragmented rather than revealing clearly distinct new groups, suggesting that the underlying data structure may not support a high number of well-separated partitions.

This behavior is even more evident in the 2D embeddings:

- for $k = 2$, a meaningful separation appears between a compact right-side cluster and a more dispersed central-left structure;
- for $k = 3-4$, an additional central cluster is identified but remains highly overlapping, indicating weak separability;
- for $k \geq 5$, no new clearly separated groups emerge, clusters overlap and are not well separated

Overall, these results suggest that the data exhibit some coarse-grained structure ($\approx 2-3$ clusters) and that increasing k mainly introduces over-segmentation rather than new informative patterns.

CLUSTERING EVALUATION AND OPTIMAL NUMBER OF CLUSTER

To quantitatively assess the K-medoids results, we evaluated clustering quality across different values of k using three standard metrics: Calinski–Harabasz (CH), Davies–Bouldin (DB), and Silhouette score. Note that CH and DB were computed on the low-dimensional embedding (as a proxy), while the Silhouette score was calculated using the original custom distance matrix.

The results show a consistent preference for low values of k . Specifically, both CH and DB indices identify $k = 3$ as optimal (maximum CH, minimum DB), while the Silhouette score peaks at $k = 2$, and then steadily decreases as k increases.

The final number of clusters was selected using a median voting strategy across the three evaluation metrics, yielding a final choice of $k = 3$. This result is coherent with visual inspection.

To further explore the intrinsic organization of the data, we applied hierarchical clustering using the same customized distance matrix. The resulting dendrogram reveals a clear multi-level structure, with three main macro-branches emerging below the chosen cut threshold. In particular, one branch appears well separated at higher linkage

distances, while the remaining observations form two substructures that merge at lower distances.

Importantly, this hierarchical view is consistent with previous findings: a strong primary split (captured by $k = 2$) and a secondary subdivision (supporting $k = 3$).

CLUSTER PROFILING AND PERSONA INTERPRETATION

To interpret the identified clusters, we assigned cluster labels to the original dataset and analyzed their characteristics through summary statistics, categorical distributions, and visual inspection (pairplots and boxplots). This allowed us to move from purely geometric clusters to interpretable profiles (personas).

- The results highlight three clearly differentiated groups. Cluster 0 is characterized by higher income and wealth, along with elevated financial literacy, digital engagement, and savings behavior, suggesting a financially advanced and proactive segment.
- Cluster 1 shows the lowest values across most financial variables (income, wealth, financial education, digital usage), combined with lower debt, indicating a more fragile or less engaged segment.
- Cluster 2 occupies an intermediate position, with moderate economic resources and behaviors, slightly larger family size, and more balanced profiles across variables, representing a middle segment with heterogeneous characteristics.

These differences are consistently visible in both the pairwise distributions and boxplots, where clusters show distinct central tendencies rather than sharp separations, confirming that segmentation captures gradual behavioral patterns rather than strictly discrete groups.

BAYESIAN ANALYSIS

We finally extended the analysis from static clustering to a dynamic persona assignment and updating framework.

First, cluster-specific financial profiles were estimated from the original clustered dataset using the mean and standard deviation of the main financial variables, together with empirical cluster priors. Then, 100 synthetic new clients were generated by bootstrap resampling with small Gaussian perturbations, and each of them was assigned to a cluster through a Bayesian classification rule, where posterior probabilities were computed as $P(C_k|x) \propto P(x|C_k) P(C_k)$. This allowed also an explicit quantification of assignment uncertainty, highlighting both highly confident cases and borderline profiles. Most new clients were assigned to Cluster 2, confirming its role as the broadest and most central

segment, while Cluster 0 remained more selective and Cluster 1 more specific. In a second step, cluster personas were updated using a multivariate Gaussian Bayesian update in logit space, so that new assigned observations could refine the prior cluster profiles and reduce uncertainty. The updated personas remained coherent with the original interpretation but became more sharply defined:

- Cluster 0 was reinforced as a high-value, digitally engaged, luxury-oriented segment, with increases in income, wealth, financial literacy, digital behavior and luxury propensity;
- Cluster 1 became even more clearly a low-engagement and less sophisticated segment, with lower income, debt, digital use, lifestyle and luxury, but slightly stronger ESG and relationship-banking orientation;
- Cluster 2 is consolidated as an intermediate and pragmatic segment, with moderate financial characteristics, higher saving and digital engagement, but lower lifestyle and luxury orientation. Importantly, all clusters showed a substantial reduction in uncertainty after updating, as reflected by the marked decrease in covariance trace, indicating that the Bayesian procedure not only adapts the personas to new evidence, but also makes them more stable and informative.

LEARNING VECTOR QUANTIZATION

In this step, we move from simple clustering to a method that allows us to learn the cluster structure and reuse it on new data. First, we generate synthetic clients for each cluster using the original cluster centroids and variability, so that the artificial data closely resemble the real segments.

Next, we train a LVQ (Learning Vector Quantization) model on the synthetic data. In simple terms, LVQ learns a few representative points (prototypes) for each cluster. LVQ models are

supervised artificial neural networks used for classification and clustering by defining prototypes in the data space. They update prototype vectors based on training data, moving them closer to correct classifications and away from errors to improve classification accuracy. Once trained, we use the model to assign cluster labels to the real data based on similarity to these prototypes.

The results show that this approach works well: the cluster structure is preserved, as confirmed by similar Silhouette (approximately 0.20, indicating a relatively low value) and Dunn index values compared to the original clustering, and a high agreement rate between the two methods. The 3D plots also visually confirm that the clusters remain well separated and consistent. Overall, this step demonstrates that we now can classifying new clients in a consistent way.

MORE WEIGHT ON AGE INTO THE CLUSTER STRUCTURE

During the analysis, we identified a key limitation in the previous clustering configuration: age was not effectively influencing the segmentation. All clusters showed relatively high average age values, with no clear distinction between younger and older clients. This produced unrealistic profiles, given that younger clients are typically less financially experienced. This issue stemmed from age being treated as a standard continuous variable and assigned a very small part of the total weight (5%).

To address this, we redefined the distance function using a two-fold approach:

1. **Age Binning:** Instead of using exact numerical ages, we converted the continuous age data into an ordinal "life stage" variable (AgeBin). By calculating tertiles based on our dataset's actual distribution, we grouped clients into three distinct, equal-sized brackets (Young, Middle, and Senior). This ensures the algorithm recognizes meaningful generational differences rather than treating a one-year age gap the exact same across the board.
2. **Weight Rebalancing:** We isolated this binned age variable into a dedicated component within the distance matrix and assigned it a higher, explicit weight. To accommodate this shift while ensuring the model's core focus doesn't change, financial variables remain dominant but were slightly adjusted to a weight of 0.78 (down from 0.80).

From a mathematical perspective, this change leads to slightly less well-separated clusters. This is visible in the 3D embedding, where clusters appear more overlapping, and is confirmed by the relatively low Silhouette score (0.1637), suggesting a limited degree of cluster separation.

However, from a practical and interpretative perspective, the updated clustering is more realistic. The increased weight of age allows the emergence of segments that better reflect real-world patterns, including a clearer distinction between younger and older clients, which was previously absent.

Overall, this modification highlights a trade-off: the previous model achieved better numerical separation but produced less meaningful segments, whereas the updated model yields slightly less compact but more realistic and interpretable clusters, better aligned with expected demographic and financial behavior.

